

ZHARKENT, KazakhstanWMO: **368590**Lat: **44.1628N**Lon: **80.0458E**Elev: **645**StdP: **93.81**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.6	-16.9	-23.3	0.5	-19.1	-20.3	0.7	-16.3	6.3	-5.9	5.1	-3.8	1.3	10	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	35.0	19.9	33.5	19.2	32.1	18.7	21.2	31.8	20.4	31.0	19.6	30.0	1.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	14.0	25.1	16.9	13.0	24.2	16.0	12.3	24.0	65.1	31.9	61.7	31.2	59.1	30.2	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	5.8	4.5	DB	-22.0	37.7	3.6	1.1	-24.7	38.5	-26.8	39.1	-28.8	39.7	-31.5	40.4	(4)
(5)			WB	-22.3	23.5	3.6	1.1	-24.9	24.3	-27.0	25.0	-29.0	25.6	-31.6	26.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	-6.3	-2.8	6.1	13.7	18.8	23.0	24.6	18.7	11.7	3.8	-3.7	(6)		
(7)		DBStd	11.59	5.15	5.15	5.28	4.61	3.76	2.93	2.66	3.01	3.77	4.44	4.88	4.68	(7)	
(8)		HDD10.0	1675	506	358	142	19	1	0	0	0	1	33	193	423	(8)	
(9)		HDD18.3	3295	764	592	379	146	39	3	1	3	41	209	436	682	(9)	
(10)		CDD10.0	2041	0	0	21	132	274	390	453	416	262	86	7	0	(10)	
(11)		CDD18.3	617	0	0	0	8	55	143	196	159	52	3	0	0	(11)	
(12)		CDH23.3	6993	0	0	5	159	648	1527	2174	1819	610	52	0	0	(12)	
(13)		CDH26.7	2992	0	0	0	38	209	644	1036	851	207	8	0	0	(13)	
(14)	Wind	WSAvg	1.8	1.4	1.6	2.1	2.5	2.5	2.1	1.9	1.8	1.7	1.6	1.5	1.3	(14)	
(15)	Precipitation	PrecAvg	180	10	9	12	21	19	27	24	11	8	14	17	14	(15)	
(16)		PrecMax	299	73	39	37	60	53	74	89	37	31	54	57	82	(16)	
(17)		PrecMin	86	0	0	0	0	0	3	0	0	0	1	0	0	(17)	
(18)		PrecStd	59	11	8	9	15	12	20	20	9	8	14	13	13	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	11.8	23.6	30.4	33.2	35.1	37.4	36.5	33.4	27.8	19.6	9.3	(19)	
(20)			MCWB	2.8	5.6	11.5	15.2	17.1	19.2	21.0	21.0	18.7	14.8	9.9	4.1	(20)	
(21)		2%	DB	3.7	8.2	20.2	27.3	31.0	33.4	35.4	34.6	30.9	24.7	16.2	6.3	(21)	
(22)			MCWB	0.6	3.4	10.1	14.3	16.6	18.7	20.2	20.0	17.5	13.5	8.8	2.1	(22)	
(23)		5%	DB	1.9	5.8	17.5	24.9	29.1	32.1	33.7	33.1	29.2	22.1	13.1	4.1	(23)	
(24)			MCWB	-0.2	2.2	9.1	13.4	15.9	18.3	19.7	19.4	16.8	12.4	7.4	1.0	(24)	
(25)		10%	DB	0.5	3.7	14.9	22.6	27.1	30.5	32.0	31.4	27.1	19.5	10.5	2.2	(25)	
(26)			MCWB	-1.1	0.9	8.1	12.4	15.0	17.8	19.1	18.8	15.9	11.3	6.1	0.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	5.8	12.7	16.7	19.2	21.2	23.0	22.4	20.1	15.9	11.6	4.9	(27)	
(28)			MCD B	5.9	10.3	21.1	26.1	29.9	31.2	33.4	33.6	30.6	25.4	16.9	8.6	(28)	
(29)		2%	WB	1.3	3.9	11.0	15.2	17.6	20.0	21.5	21.0	18.4	14.3	9.4	2.7	(29)	
(30)			MCD B	2.9	7.6	18.4	25.1	28.0	30.3	31.6	32.5	29.2	22.9	14.8	5.5	(30)	
(31)		5%	WB	0.1	2.4	9.6	14.0	16.7	19.3	20.6	20.1	17.3	13.0	8.0	1.3	(31)	
(32)			MCD B	1.6	5.2	16.4	23.3	26.2	29.2	31.0	31.1	27.8	20.2	12.7	3.7	(32)	
(33)		10%	WB	-1.1	1.2	8.4	13.0	15.8	18.6	19.9	19.3	16.3	11.9	6.5	0.1	(33)	
(34)			MCD B	0.6	3.5	14.2	21.0	24.9	28.0	30.2	29.8	25.9	18.3	10.1	2.0	(34)	
(35)	Mean Daily Temperature Range	MDBR		9.8	9.4	10.9	12.7	12.9	12.6	13.0	14.0	14.2	12.7	9.9	9.1	(35)	
(36)		5% DB	MCD BR	10.9	11.0	14.9	16.5	15.6	15.0	15.6	16.4	17.0	16.2	13.9	11.5	(36)	
(37)			MCWBR	7.9	6.9	7.1	7.6	6.1	5.5	6.1	7.3	8.1	7.7	7.7	7.9	(37)	
(38)		5% WB	MCD BR	9.2	9.8	13.2	14.3	13.6	12.8	13.6	15.0	16.0	13.8	12.7	10.4	(38)	
(39)			MCWBR	6.8	6.3	6.6	6.8	5.8	5.2	5.8	7.0	7.9	6.8	7.3	7.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.315	0.358	0.413	0.447	0.422	0.420	0.412	0.388	0.373	0.372	0.349	0.313	(40)	
(41)		taud		2.213	2.119	2.014	1.985	2.135	2.209	2.252	2.304	2.292	2.196	2.172	2.209	(41)	
(42)		Ebn at Noon		784	807	808	814	850	852	854	864	846	783	728	742	(42)	
(43)		Edh at Noon		90	119	151	169	150	140	133	122	113	109	94	82	(43)	
(44)	All-Sky Solar Radiation	RadAvg		2.04	3.01	3.96	5.11	6.31	6.53	6.49	6.04	4.88	3.13	1.95	1.63	(44)	
(45)		RadStd		0.11	0.16	0.31	0.41	0.37	0.30	0.40	0.32	0.27	0.31	0.17	0.12	(45)	